1 Model

$$\|\eta\|_*^2 = \sum_{s<L} \|\eta^s\|^2/N + |\eta^L|^2. \tag{1}$$

2 Robust posterior compression uniformly for $\gamma \geq 1$

This section gives a different statistical bridge. It uses robustness of the original network while constructing a finite candidate, and tracks entropy only in the original weights that have not yet been processed. Consequently the intermediate compressed prefix need not have a density under a Gaussian prior. The result applies to a posterior containing many different functions.

Let $\mu_{N,t}$, $0 < t \leq 1$, have independent weight-entry variances t^2/d, t^2/N, and t^2/N^2 in the first, internal, and readout layers. The exponent-γ prior is obtained by $t = N^{(1-\gamma)/2}$. All-layer sign robustness refers to the total noise norm in (1); the clean 1/2 margin may additionally be imposed.

Theorem 2.1 (Uniform robust posterior compression)**.** *Fix $d \geq 1$, $L \geq 2$, $K \geq 1$, $0 < r \leq 1$, and $0 < \alpha \leq 1/8$. There is a finite class $\mathcal{H} = \mathcal{H}(d, L, K, r, \alpha)$, consisting of signs of bounded-weight depth-L networks, such that*

$$\log |\mathcal{H}| \leq \mathrm{poly}(d, L, K, r^{-1}, \alpha^{-1}).$$

The polynomial is universal and does not depend on N or t. If $Q \ll \mu_{N,t}$ is supported on all-layer sign-r robust networks and

$$D_{\mathrm{KL}}(Q\|\mu_{N,t}) \leq NK/t^2,$$

then there is a distribution Γ_Q on $\mathcal{H}$ such that, for every $x \in \{-1,1\}^d$, a coupling of $w \sim Q$ and $h \sim \Gamma_Q$ satisfies

$$\mathbb{P}\{h(x) \neq \operatorname{sign} z_w(x)\} \leq \alpha.$$

The coupling may depend on x; its two marginals do not. Neither the support of Q nor its samples need represent a single Boolean function.

We give the proof, including deliberately loose polynomial choices. Two elementary entropy facts organize the construction. First, for a standard D-dimensional Gaussian G, the entropy variational inequality gives

$$\mathbb{E}_P\|Z\|^2 \leq 4D_{\mathrm{KL}}(P\|G) + 2D\log 2. \tag{2}$$

Second, Gaussian Talagrand gives $W_2^2(P,G) \leq 2D_{\mathrm{KL}}(P\|G)$.

Lemma 2.2 (Entropy of the unprocessed weights)**.** *Let A, X, Y be random variables, with fixed reference laws P_X, P_Y, and put*

$$B = \mathbb{E}_A D_{\mathrm{KL}}(Q_{X,Y|A}\|P_X \otimes P_Y).$$

Let J be a finite message, possibly depending on (A, X, Y), and let $H = f(A, J, X)$. If A' is a randomized function of (A, J, H) whose fresh randomness is conditionally independent of Y, then

$$\mathbb{E}_{A'} D_{\mathrm{KL}}(Q_{Y|A'}\|P_Y) \leq B + H(J \mid A).$$

Proof. The left side equals $D_{\mathrm{KL}}(Q_Y\|P_Y) + I(Y; A')$. Data processing and

$$I(Y; J, H \mid A) \leq I(Y; X, J \mid A) \leq I(Y; X \mid A) + H(J \mid A)$$

bound it by $\mathbb{E}_{A,X} D_{\mathrm{KL}}(Q_{Y|A,X}\|P_Y) + H(J \mid A)$, which is at most the claimed bound by the KL chain rule. These expressions can equivalently be justified by finite partitions and lower semicontinuity. □

Here A will be a constructed prefix, X the current original weight block, and Y all later original blocks. The lemma imposes no reference law on A.

Lemma 2.3 (Whole-layer active directions)**.** *Condition on (A, Y) and let X be a standardized Gaussian-reference matrix, with conditional relative entropy $R(A, Y)$. For vectors $b_x(A)$ of norm at most one, one can select a list J of inputs and its span $U = \operatorname{span}\{b_x : x \in J\}$ with*

$$\dim U \le \frac{t^2 R(A, Y)}{N\delta}$$

such that preserving $H = tXU$ and replacing the orthogonal coordinates by fresh Gaussian coordinates changes the physical evaluation vectors tXb_x, under a suitable coupling, by expected squared RMS at most 2δ at each x. The candidate matrix law preserves (A, Y, H) and does not depend on x.

Proof. Use a fixed ordering of the finite input set. Whenever the averaged conditional KL of the entire residual evaluation vector Xv exceeds $N\delta/t^2$, add the unit residual direction v of that input. Orthogonal Gaussian coordinates are independent, so the active-coordinate entropy increases by precisely this conditional KL. The displayed rank bound follows. At termination, Gaussian Talagrand on the entire evaluation vector, multiplication by the squared residual-vector length, and division by N/t^2 give the error bound. Scalar-vector couplings extend to full matrices by disintegration. The basis of U is a measurable function of A, J under fixed Gram–Schmidt; it has no further dependence on Y once those variables are supplied. □

Parameter choices. Write

$$\begin{aligned}
e &= \alpha r^2/(1000L), & \eta &= \alpha/(100L), & q &= \lceil 2K/(e\eta)\rceil,\\
D_0 &= [(d+1)q + 2]\log 2, & V &= 8K + 2q, & B_H^2 &= 4V/e,\\
T &= 128(q+1)(B_H^2+1)/e, & a &= e/[128(B_H^2+1)],\\
K_0 &= (4K + 2\log 2)/\eta, & B_v &= 16K_0/r.
\end{aligned}$$

Choose m at least

$$\begin{aligned}
&10000T^4(q+1)^2a^{-2}[d + \log(10(q+1)/\eta) + 1],\\
&10000B_v^2 r^{-2}[d + \log(10/\eta) + 1],
\end{aligned}$$

and put

$$N_0 = \max\{m, \lceil LD_0/K\rceil, 1\}, \quad B_G = \max\{2, 2\sqrt{\log(3m/e)}\}, \quad B_1^2 = 4[4K + 2(d+1)\log 2]/e.$$

All quantities have a fixed polynomial upper bound in the theorem parameters.

The small-width case. If $N \le N_0$, let $P_0 = N_0(d+1) + (L-2)N_0^2 + N_0$. Applying (2) to all standardized weights, and then multiplying by t^2, gives

$$\mathbb{E}_Q\|tZ_{\text{all}}\|^2 \le 4N_0K + 2P_0\log 2.$$

With probability at least $1 - \alpha/4$, all actual weights are bounded by

$$B_{\text{small}} = [4(4N_0K + 2P_0\log 2)/\alpha]^{1/2}.$$

Keep the original network on this event and use a default classifier otherwise. Padding with nodes having zero outgoing weights preserves the original output. The successful candidate has output margin greater than r. It remains to treat $N > N_0$.

The virtual prefix and its entropy budget. The constructed prefix A has N virtual nodes in its current layer. Each virtual row acts on m stored nodes of the preceding layer. The auxiliary description may keep all N rows; the final network will keep just the rows selected at the following stage. Track the conditional KL of the unprocessed *original* weights. By Lemma 2.2, it is at most

$$B_s \le NK/t^2 + sD_0 \le 2NK/t^2. \tag{*}$$

This is not a KL bound for the singular compressed prefix itself.

For the first layer, zero every physical standardized row $tZ_{1,i}$ of norm greater than B_1, and retain all other rows. Bounded activations and (2) bound the expected squared RMS activation change at each input by e. This deterministic prefix channel does not increase the future entropy budget.

An internal layer: retain active data. Condition on the current prefix A and the future original weights Y. Apply Lemma 2.3 to $b_x = h_A(x)/\sqrt{N}$ and $\delta = e$. Abort if more than q inputs are needed. By (*) and Markov, this failure has probability at most η. The successful query list, together with an abort flag, has a code of at most $[(d+1)q+2]$ bits. Its entropy is at most D_0.

Let $H = tXU$ be the physical active matrix. Averaging the Gaussian entropy moment bound, with the success indicator rather than conditioning on success, gives

$$\mathbb{E}\left[\frac{\|H\|_F^2}{N};\ \text{success}\right] \le 4t^2 B_s/N + 2t^2 q \log 2 \le V.$$

Zero every row H_i whose norm exceeds B_H. The expected squared RMS activation change from this operation is at most $4V/B_H^2 = e$. All rows are clipped before any later selection, including selections that may depend on future weights.

An internal layer: reconstruct using m source nodes. For the orthonormal active basis let $u(j) = \sqrt{N}\, U_{j,:}$; uniform source sampling gives $\mathbb{E}_j u(j)u(j)^\top = I$. Clip the *whole vector* at radius T: $c(j) = u(j)\mathbf{1}_{\{\|u(j)\| \le T\}}$. Then

$$\mathbb{E}cc^\top \preceq I, \qquad \mathbb{E}\|u - c\| \le q/T.$$

Sample m uniform source indices S_j. Entrywise Hoeffding and a union bound give, with probability at least $1-\eta$,

$$\|\widehat{G} - \mathbb{E}cc^\top\|_{\mathrm{op}} \le a, \quad \sup_x \|\widehat{v}(x) - \mathbb{E}ch_A(x)\| \le a,$$
$$\sup_{x,y} |\widehat{K}(x,y) - K(x,y)| \le a,$$

where $\widehat{G} = m^{-1}\sum_j c(S_j)c(S_j)^\top$, $\widehat{v}(x) = m^{-1}\sum_j c(S_j)h_{A,S_j}(x)$, and $\widehat{K}$ is the empirical feature covariance. Abort on failure; this event depends only on A, J, S, so it does not reveal further future information.

Set $\widetilde{c}_j = c(S_j)/\sqrt{1+a}$, and let D have rows $\widetilde{c}_j^\top/\sqrt{m}$. Then $D^\top D \preceq I$ and $I - DD^\top \succeq 0$. Independently for every target virtual row draw $\xi_i \sim N(0, I - DD^\top)$ and use coefficients

$$\widehat{w}_{ij} = m^{-1} H_i^{\mathrm{clip}} \cdot \widetilde{c}_j + t\xi_{ij}/\sqrt{m}. \tag{3}$$

These are actual neural-network rows on the m sampled nodes. Their active feature vector is $\widehat{\psi}(x) = m^{-1}\sum_j \widetilde{c}_j h_{A,S_j}(x)$, and the lazy covariance is exactly $t^2(\widehat{K} - \widehat{\psi}^\top\widehat{\psi})$. For the ideal field, write $\psi(x) = U^\top b_x$ and $C(x,y) = K(x,y) - \psi(x)^\top\psi(y)$. Direct calculation gives

$$\sup_x \|\widehat{\psi}(x) - \psi(x)\| \le q/T + 2a =: \zeta, \qquad \sup_{x,y} |\widehat{C}(x,y) - C(x,y)| \le a + 2\zeta.$$

At a fixed input, coupling scalar Gaussian evaluations with a common standard normal bounds the squared preactivation transport error, per row, by

$$B_H^2\zeta^2 + t^2(a + 2\zeta) \le e/8.$$

Extend these scalar couplings to full row fields. The candidate weight law is the same at every input; there is no claim of a single Gaussian-field coupling that is close at all inputs simultaneously.

Finally clip *all* coordinates ξ_{ij} to $[-B_G, B_G]$ before later row selection. Since each has variance at most one,

$$\mathbb{E}\left|\frac{t}{\sqrt{m}}\sum_j(\xi_{ij} - \mathrm{clip}_{B_G}\xi_{ij})h_{A,S_j}(x)\right|^2 \le 3me^{-B_G^2/4} \le e.$$

Thus every virtual coefficient is bounded by $B_H T/m + B_G/\sqrt{m}$, without a union bound over N.

All newly constructed data are a randomized function of A, J, H. Lemma 2.2 therefore proves (*) for the next step. Combining entropy replacement, active clipping, finite reconstruction, and Gaussian clipping by $(a_1 + \cdots + a_4)^2 \le 4\sum_i a_i^2$, the expected squared RMS activation injection at this layer is less than $20e$.

The readout and final network. The original readout marginal has never changed. Its physical moment satisfies $\mathbb{E}_Q[N\|v\|^2] \le 4K + 2\log 2$. Abort if $N\|v\|^2 > K_0$, an event of probability at most η. Zero coefficients with $|Nv_i| > B_v$. This changes the output uniformly on bounded virtual feature vectors by at most $K_0/B_v = r/16$. Sampling m uniform last-layer sources and using readout coefficients $(Nv_i\mathbf{1}_{|Nv_i|\le B_v})/m$ adds at most $r/16$ uniformly on $\{-1,1\}^d$, except with probability η, by Hoeffding and the chosen m.

At each stage, retain in the final network just the m source rows selected from the preceding virtual layer. Each such row already acts on the previously stored m nodes. The result is an ordinary width-m network; repeated indices are duplicate nodes. Its weights are bounded by

$$B_{\text{large}} = \max\{B_1, B_H T/m + B_G/\sqrt{m}, B_v/m, 1\}.$$

The larger virtual construction is not part of the final predictor.

The original network receives all perturbations. Fix an input x and glue the conditional couplings, retaining the original weight history. The current and future original blocks are untouched until their own stages. Define

$$\eta^s = h^s_{\text{virtual}}(x) - \phi(W^s_{\text{original}}h^{s-1}_{\text{virtual}}(x)).$$

The analogous readout difference is η^L. Later stages leave earlier computational prefixes unchanged, so the final candidate output is exactly the *original* network's output under these postactivation injections. No intermediate robustness assumption is made.

For precision, the canonical construction draws its new Gaussian coefficients independently of the discarded current block given (A, J, H). The comparison coupling may correlate them with that discarded block. It preserves the same joint marginal of the new prefix and the future original blocks, as well as the original full-network marginal, but need not preserve the canonical joint law with all discarded weights. The next entropy step uses the first marginal; original-network robustness uses the second. There is no conflicting demand for independence in the comparison coupling itself.

Abort probability is at most $2L\eta$. The unconditional moment bound gives

$$\mathbb{P}\left\{\text{no abort and } \sum_{s<L} \|\eta^s\|^2/N > r^2/8\right\} \le 160Le/r^2 = 0.16\alpha.$$

Otherwise the readout injection is at most $r/8$, and total squared noise is at most $9r^2/64 < r^2/4$. Robustness then gives candidate margin greater than $r/2$ relative to the original network's label: a smaller margin could be cancelled by an additional output shift within the radius-r noise ball. The exceptional probability is less than $\alpha/4$.

A fixed finite class. Pad candidates to width $M = N_0$ and take $B = \max(B_{\text{small}}, B_{\text{large}}, 1)$. Round every parameter to a fixed grid on $[-B, B]$ of mesh at most

$$h = \frac{r}{16(d+L+1)(2M(B+1))^L}.$$

The bounded-activation, Lipschitz layer recursion gives uniform output change at most $r/8$. Thus rounding preserves every successful coupled prediction. Writing $P = M(d+1) + (L-2)M^2 + M$, the grid class satisfies

$$\log|\mathcal{H}| \le P \log\left[3 + \frac{32B(d+L+1)(2M(B+1))^L}{r}\right].$$

Although grid precision is exponentially small in L, its number of bits is polynomial in L. All other bounds above are polynomial too. Rounding is a fixed map on candidate networks, giving the common marginal Γ_Q and proving Theorem 2.1.

Corollary 2.4 (Polynomial-depth learning at every width)**.** *Suppose the raw Gaussian target event, consisting of networks with all-layer sign margin r and clean margin $1/2$ for f, has mass at least $\exp(-NK/t^2)$. Restrict the prior to all sign-r robust networks having global clean margin $1/2$, regardless of their Boolean function, then condition on n noiseless iid training examples. With probability at least $1-\beta$, the resulting posterior satisfies*

$$\mathrm{GibbsErr}(Q) \le 2\alpha + \sqrt{\frac{\log(2|\mathcal{H}|/\beta)}{2n}}.$$

Consequently the sample bound is jointly polynomial in d, L, K, r^{-1} and inverse desired error, uniformly in N and $0 < t \le 1$. For posterior-confidence error at threshold $2/3$, multiply Gibbs error by three; for threshold $1/2$, multiply it by two.

Proof. The posterior is the raw Gaussian conditioned on an event containing the target event, so its KL is at most NK/t^2. Its training error is zero. The compression couplings imply that Γ_Q has expected empirical error at most α. Finite-class uniform convergence holds for every member of $\mathcal{H}$, hence also for this sample-dependent mixture. Its true error is at most α plus the displayed square-root term. The pointwise coupling adds at most one more α to recover original posterior Gibbs error. This argument does not require a global coupling over inputs or a common function for all draws. The confidence-error statements follow from Markov on posterior error probability at a random test input. □